Lec 14

Recap

Joint PMF $P_{X,Y}(x,y) := P(\{X=x\} \cap \{Y=y\})$

Marginal PMF $P_X(x) = \sum_{y \in \mathcal{Y}} P_{X,Y}(x,y)$, $\sum_{x,y} P_{X,Y}(x,y) = 1$

Functions of two R.Vs.

$$Z = g(X,Y)$$

$$P_Z(z) = \sum_{(x,y):\, g(x,y)=z} P_{X,Y}(x,y)$$

$$E[Z] = \sum_{(x,y)} g(x,y)\, P_{X,Y}(x,y)$$

$$Y = g(X)$$

$$P_Y(y) = \sum_{x:\, g(x)=y} P_X(x)$$

$$E[Y] = \sum_{x \in \mathcal{X}_e} g(x)\, P_X(x)$$

$$E[a_1X_1 + a_2X_2] = a_1E[X_1] + a_2E[X_2]$$

Altenate Proof : Assume Ω is countable

$$E[X] = \sum_{\omega\in\Omega} X(\omega)\,P(\{\omega\})$$

$$E[X] = E[a_1X_1 + a_2X_2] = \sum_{\omega\in\Omega} (a_1X_1 + a_2X_2)(\omega)\,P(\{\omega\})$$

$X = a_1X_1 + a_2X_2$

$$= \sum_{\omega\in\Omega} \left(a_1X_1(\omega) + a_2X_2(\omega)\right)P(\{\omega\})$$

$$= a_1E[X_1] + a_2E[X_2]$$

Let $X = g(X_1, X_2, X_3, \ldots X_n)$

$$P_{X_1, \ldots X_n}(x_1, \ldots x_n) = P(\{X_1 = x_1\} \cap \cdots \{X_n = x_n\})$$

$$P_X(x) = \sum_{\substack{(x_1, \ldots x_n) \text{ st} \\ g(x_1, \ldots x_n) = x}} P_{X_1, \ldots X_n}(x_1, \ldots x_n)$$

$$E[X] = \sum_{(x_1, \ldots x_n)} g(x_1, x_2, \ldots x_n) P_{X_1, \ldots X_n}(x_1, \ldots x_n)$$

Linearity of Expectation

$$E\left[a_1 X_1 + a_2 X_2 + \cdots a_n X_n\right] = \sum_{i=1}^{n} a_i E[X_i]$$

Examples

① $X \sim \text{Binomial}(n, p)$

$X_i \sim \text{Bernoulli}(p)$.

$$X_i(\omega) = \begin{cases} 1 & \omega_i = H \\ 0 & \omega_i = T \end{cases}$$

$\omega = (\omega_1 \omega_2 \dots \omega_n)$

$\Omega = \{H, T\}^n$.

$X(\omega) = \#$ of heads in ω.

$\Rightarrow X(\omega) = X_1(\omega) + \dots X_n(\omega)$

$\Rightarrow E[X] = \sum_{i=1}^{n} E[X_i] = np$.

② Negative Binomial (p, r).

\# of tosses needed to see r heads.

r=1

$$\Omega_1 = \{H, TH, TTH, \ldots\}$$

r=2

$$\Omega_2 = \{HH, HTH, HTTH, \ldots\} = \{H\} \otimes \Omega_1$$

$$\cup \{THH, THTH, THTTH, \ldots\} = \{TH\} \otimes \Omega_1$$

$$\cup \{TTH|H, TTH|TH, TTH|TTH, \ldots\}$$

$$\vdots$$

$$\Omega_2 = \Omega_1 \times \Omega_1$$

$X(\omega)$ = length of ω.

$X_1(\omega)$ = # of tosses untill first head is spotted.

$X_1(\{TTHH\}) = 3.$

$X_2(\omega) = \underbrace{\text{\# of tosses done till you second head} - \text{\# of tosses done till the first head}}_{\text{\# of extra tosses after first head to see the seco}}$

$X_2(\{TTHH\}) = 1$

For general r, $i \le r$.

$X_i(\omega)$ = # of tosses done after the $(i-1)$th head is seen until ith head.

$$X(\omega) = X_1(\omega) + \cdots X_r(\omega)$$

$X_1 \sim \text{Geometric}(p)$

$X_i \sim \text{Geometric}(p)$

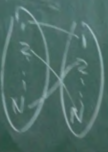
$$E[X] = \sum_{i=1}^{r} E[X_i]$$

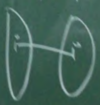
$$= \frac{r}{p}$$

(3) Consider the problem of matching hats where N people put in their hats at the center and pick a hat randomly without replacement. Let X be a r.v representing the expected # of matches

E_i = i-th person gets their matching hat

$$X_i(\omega) = \begin{cases} 1 & \omega \in E_i \\ 0 & \omega \notin E_i \end{cases}$$





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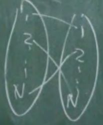

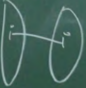

$$P(X_i = 1) = P(E_i)$$

$$E[X_i] = P(X_i = 1) \times 1 + 0 \times P(X_i = 0)$$
$$= P(E_i)$$

$$P(E_i) = \frac{(N-1)!}{N!} = \frac{1}{N}$$

$$\Rightarrow E[X_i] = \frac{1}{N}$$

$$X = X_1 + X_2 + \cdots X_N$$

$$E[X] = \sum_{i=1}^{N} E[X_i] = N \times \frac{1}{N} = 1$$

④ N different types of coupons & each time one obtains a coupon it is equally likely to be any of these N types. Find the expected # of coupons one amasses before obtaining the complete set of coupons.

X = # of coupons to be collected

$X_1 = 1$

X_2 = # of coupons collected after first coupon until you second distinct coupon

$X_2 \sim \text{Geomet}\left(\frac{N-1}{N}\right)$

w.p $\left(\frac{N-1}{N}\right)$ you'll get distinct

w.p $\frac{1}{N}$ you'll collect the same coupon you already have

$1 \quad \underbrace{1 \xrightarrow{1/N} 1}_{} \xrightarrow{\frac{N-1}{N}}$

$$X = X_1 + X_2 + \cdots X_N$$

$$X_i \sim \text{Geometric}\left(\frac{N-i+1}{N}\right)$$

$$E[X_i] = \frac{N}{N-i+1}$$

$$E[X] = \sum_{i=1}^{N} \left(\frac{N}{N-i+1}\right)$$

$$= N \sum_{i=1}^{N} \left(\frac{1}{N-i+1}\right)$$

Conditioning of a R.V X

(a) Given an event A s.t $P(A) > 0$

$$P_{X|A}(x) = P(\{X=x\} | A) = \frac{P(\{X=x\} \cap A)}{P(A)}$$

Example: $\Omega = \{1, 2, \dots 6\}$

A = event that the outcome is even.

$X(\omega) = \omega$, $P(\{\omega\}) = 1/6$.

$P_X(x) = 1/6 \quad \forall x \in \{1, 2, \dots 6\}$.

$$P(A) = P(\{2, 4, 6\}) = 3/6 = 1/2.$$

$P_{X|A}(x)$

(b) Giver

$P_{X|Y}($

given $P(\{Y=y$

(Condit

X

st $P(A) > 0$

$=x\} / A)$

$x\} \cap A)$

$)$

me is even

$1/6$

$2, \ldots 6\}$

$3/6 = 1/2$

$$P_{X|A}(x) = \frac{P(\{X=x\} \cap A)}{P(A)}$$

$$= \begin{cases} 0 & x \notin A \\ \dfrac{P(\{x\})}{P(A)} = \dfrac{1/6}{1/2} = \dfrac{1}{3} & x \in A \end{cases}$$

(b) Given a random variable Y.

$$P_{X|Y}(x|y) = P_{X|\{Y=y\}}(x)$$

given $P(\{Y=y\}) > 0$

(Conditional PMF)

$P_{X|Y}(x|y)$

$P_{Y|X}(y|x$

① $\sum_{y} P_{Y|X}$

② $\sum_{x} P_{X}$

$$P_{X|Y}(x|y) = \frac{P(\{X=x\}\cap\{Y=y\})}{P(\{Y=y\})}$$

$$= \frac{P_{X,Y}(x,y)}{P_Y(y)}$$

$$P_{Y|X}(y|x) = \frac{P_{X,Y}(x,y)}{P_X(x)}$$

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① $\sum_{y} P_{Y|X}(y|x) = 1$

② $\sum_{x} P_{X|Y}(x|y) = 1$

Multiplication rule

$$P_{X,Y}(x,y) = P_X(x)\, P_{Y|X}(y|x)$$
$$= P_Y(y)\, P_{X|Y}(x|y)$$

$$P_{X_1,X_2,\ldots X_n}(x_1,x_2,\ldots x_n) = P_{X_1}(x_1)\, P_{X_2|X_1}(x_2|x_1)\, P_{X_3|X_1 X_2}(x_3|x_1,x_2) \cdots P_{X_n|X_1,X_2,\ldots X_{n-1}}(x_n|x_1,\ldots x_{n-1})$$

$$P_{X_i|X_1,\ldots X_{i-1}}(x_i|x_1,x_2,\ldots x_{i-1}) = P_{X_i|\boxed{\{X_1=x_1\}\cap\{X_2=x_2\}\cap\ldots\{X_{i-1}=x_{i-1}\}}}(x_i)$$